



Rakesh Hadne Sreenath

AI Engineer | LLM & Agentic Systems | MLOps

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience designing and deploying production-grade LLM architectures, Agentic AI systems, and MLOps pipelines. Published researcher (ICLR 2025, bioRxiv) with proven ability to translate complex AI concepts into scalable, business-ready solutions. Currently employed at e.Ray Europa GmbH (Darmstadt). Full right to work in Germany, no sponsorship required. Fluent English (C1); German A2, actively pursuing B1.

PUBLICATIONS

Talk2BioModels | *bioRxiv* (2025)

Agentic LLM system for kinetic biological models.

Business Impact: Reduced hallucination rates by 20% in complex reasoning tasks

Talk2KnowledgeGraph | *ICLR Workshop* (2025)

Agent-based AI for biomarker prediction and reasoning.

Business Impact: Improved data retrieval accuracy by 15% using graph-based reasoning

TECHNICAL SKILLS

LLM & Agents	LangGraph, Pydantic AI, LangChain, LangSmith, Multi-Agent Orchestration, Tool/Function Calling, Prompt Engineering, RAG, MCP Servers, Cognitive Architectures
ML & NLP	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Fine-tuning (LoRA), RLHF, Semantic Search, NLP, LLM Evaluation
Cloud & MLOps	GCP (Vertex AI, BigQuery, Cloud Run), AWS (SageMaker, Lambda, S3), Azure, Docker, Kubernetes, CI/CD (GitHub Actions, GitLab), Terraform, FastAPI
Data Systems	PostgreSQL, Neo4j (Knowledge Graphs), Qdrant, Pinecone, Milvus, BigQuery, ETL/ELT Pipelines, REST/GraphQL APIs
Languages	Python (Expert), R, SQL, Bash

PROFESSIONAL EXPERIENCE

AI Engineer (Intern) | e.Ray Europa GmbH

Mar 2026 – Present | Darmstadt, Germany

- Architected an Agentic AI system that automated **80% of unstructured data processing**, reducing manual compliance overhead by **~12 hours per week**
- Optimized domain-specific LLMs through fine-tuning, achieving a **15%** increase in extraction accuracy for complex regulatory schemas
- Fine-tuned LLMs on domain-specific GDPR-compliant data ingestion pipelines for environmental workflows, ensuring **100% data residency adherence**.
- Developed a RAG-based analytics layer that accelerated historical trend analysis by 40% using PostgreSQL-backed datasets.

GenAI Engineer (Master's Thesis) | BioMed X Institute

Apr 2025 – Sep 2025 | Heidelberg, Germany

- Led design and implementation of agentic workflows for systems biology; conducted requirements analysis and feasibility studies for complex scientific pipelines.
- Built custom tools using **Pydantic AI and LangGraph**, enabling seamless LLM interaction with external biological simulation frameworks.
- Reduced CI/CD cache access time by 5+ minutes per run** and cut environment setup time to under 30 seconds via Conda-based workflow migration.

- ▶ Established LLMOps environments to monitor AI performance, hallucination rates, and ensure production-grade reliability.
- ▶ Implemented full unit test coverage across Windows, Ubuntu, and macOS, ensuring deterministic, production-ready releases.

GenAI Engineer (Intern) | BioMed X Institute

Nov 2024 – Mar 2025 | Heidelberg, Germany

- ▶ Developed Talk2BioModels: an agentic AI system enabling natural language interaction with large-scale biomedical repositories using dynamic tool calling and structured LLM APIs.
- ▶ Built robust RAG pipelines featuring episodic memory and semantic caching, improving retrieval precision and reducing operational costs.
- ▶ Evaluated and documented multiple LLM architectures for domain-specific tasks; integrated prompt engineering, NLP pipelines, and knowledge graphs.

Software Engineer | Tech Mahindra

Aug 2021 – Jun 2023 | Bengaluru, India

- ▶ Automated 7+ manual business workflows using Python and Perl, achieving 18% improvement in operational efficiency and 40% reduction in manual effort.
- ▶ Developed and integrated secure REST/GraphQL API connectors; optimized mission-critical PostgreSQL schemas, improving query performance by 18%.
- ▶ Maintained 99%+ system stability through structured regression and performance testing; resolved 30+ software defects via collaborative code review.
- ▶ Awarded 'Pat on Back' (2022) and 'Best Team Award' (2022) for consistent delivery and exceptional team collaboration.

KEY PROJECTS

Agentic Student Assistant

- ▶ Modular multi-agent academic companion using LangGraph & Pydantic AI; integrated external market data via structured APIs with full audit trails. Stack: Python, LangGraph, Pydantic AI, Vector DBs, OpenAI API.

Voice-Enabled Audio RAG

- ▶ Production-ready voice interaction system using Whisper STT and Qdrant for natural language document interrogation; included offline evaluation benchmarks for latency and accuracy. Stack: Python, Whisper, Qdrant, RAG, Docker.

Production Cloud LLMOps Pipeline

- ▶ End-to-end containerized AI pipelines using Docker and Terraform on GCP; automated monitoring dashboards for real-time health, failures, and data drift. Stack: Terraform, Docker, GCP, BigQuery.

EDUCATION

M.Sc. Applied Data Science & Analytics | SRH Hochschule Heidelberg

Oct 2023 – Sep 2025 | Germany

Focus: NLP, Agentic AI, Big Data Engineering, Statistical Modeling, Python/R

Post Graduate Program in Data Science | Purdue University

Aug 2022 – Mar 2023 | USA (Online)

Focus: Machine Learning, Predictive Analytics, Power BI, Python/R

B.E. Electronics & Communications Engineering | UBDT College of Engineering

Aug 2016 – Jun 2020 | India

LANGUAGES & WORK STATUS

English: C1 – Professional	German: A2 (pursuing B1)	Work Status: Full right to work in Germany
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